EMPLACE: Self-Supervised Urban Scene Change Detection
Supplementary Material

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Cut-and-Flip	Loss	SI-1		SI-2		SI-3		SI-4		SI-Hard	
		Acc	Epoch	Acc	Epoch	Acc	Epoch	Acc	Epoch	Acc	Epoch
✓	Adaptive	.975	13	.902	11	.937	15	.951	13	.928	16
✗	Adaptive	.965	14	.870	17	.926	14	.927	12	.888	21
✓	Fixed	.954	18	.875	26	.929	17	.948	17	.908	22

Table 1: Ablation of cut-and-flip augmentation and adaptive triplet loss. Fixed margin set to $\alpha = 1$. SI-Hard parameters are $90 < \Delta_{AP} < 365$ and $365 < \Delta_{AN}$.

Appendix

Ablation Study

We evaluate the utility of both cut-and-flip augmentation as well as the adaptive triplet loss introduced in Eq. 6. The results are shown in Table 1. We see that cut-and-flip augmentation provides an increase in performance over all setups. We also see that the adaptive margin has increased utility on setups where the negative and positive are closer together such as SI-2. We also observe that in all instances where the model trains, the adaptive margin takes the fewest number of epochs to converge. As such we believe the temporal triplet loss outperforms the fixed margin by performing similar or better in all tested setups.

	SI-1	SI-2	SI-3	SI-4	All
SI-1	**.013**	.040	.065	.099	.055
SI-2	.014	**.022**	**.024**	**.010**	**.015**
SI-3	.031	.037	.026	.017	.018
SI-4	.030	.048	.037	.033	.041
Baseline	.081	.028	.054	.083	.510

Table 2: Standard deviation over the accuracy distribution over the 8 geographical areas (stadsdelen) of Amsterdam.

Bias study

To enable learning in a self-supervised manner we build on the assumption that temporal change is proportional to visual/urban change. However, this assumption may not strictly hold in every case, as there may be certain places in the city that hardly change over the years. As such, this can potentially cause our model to be biased towards places that exhibit more visual change. To study possible bias as a result of this assumption we repeat the order prediction experiment in Table 3 split by the 8 geographical areas (stadsdelen) of Amsterdam. The values in this table represent the standard deviation of the distribution of accuracies over the spatial regions. While the variance between the accuracies over different regions is present, we see that overall this variance is larger in easier setups. Harder setups, such as SI-2, show less variance over the different regions. From this we conclude that baseline methods show strong spatial biases and that training with EMPLACE shows a significant reduction in this bias.

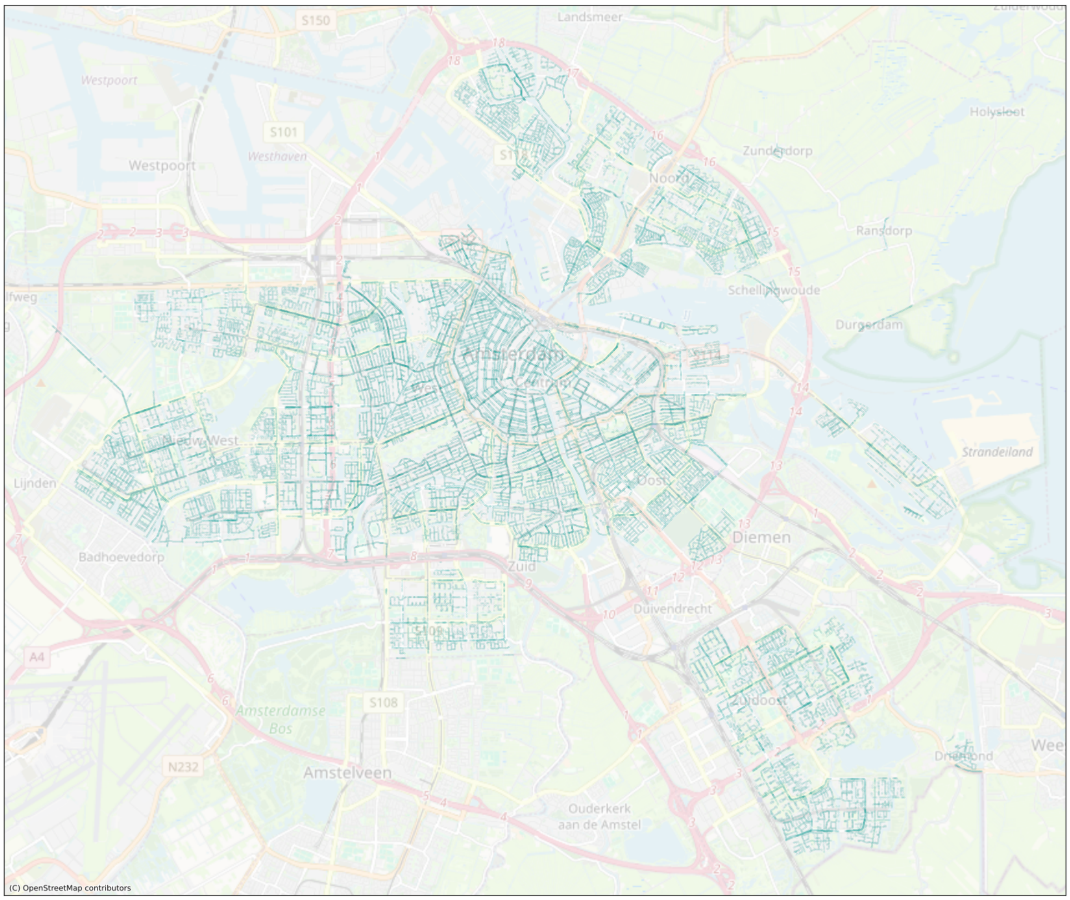

Figure 1: Plot of the 254k clusters of the AC-1M.

Figure 2: Detected small changes in Amsterdam with zero-shot EMPLACE. For each set of images, change was detected between image 2 and 3.

Figure 3: Detected large changes in Amsterdam with zero-shot EMPLACE.